UU-B-43D
July 31, 1972
SUPERSEDING
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July 23, 1964

FEDERAL SPECIFICATION

BAG, PAPER, WASTE-RECEPTACLE

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers waxed paper bags suitable for use as an insert in sanitary waste receptacles for the disposal of garbage, refuse, and infectious waste.

1.2 Classification.

1.2.1 Types. The bags covered by this specification shall be of the following types as specified (see 6.1).

Type I - Bags, Square (see Fig. 1).
II - Bags, Self-Opening (automatic) (see figs. 2 and 3).

1.2.2 Dimensions. The bags shall be of the dimensions and capacities listed in tables I and II (see 3.5.1, 3.5.2 and 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards, of the issues in effect on the date of invitation for bids, form a part of this specification to the extent specified herein.

Federal Specifications:

PPP-B-585 - Boxes, Wood, Wirebound.
PPP-B-591 - Boxes, Fiberboard, Wood-Cleated.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Box, Shipping, Fiberboard.
PPP-B-640 - Boxes, Folding, Fiberboard, Corrugated, Triple-Wall.

Federal Standard:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specification and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks, and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection By Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

Technical Association of the Pulp and Paper Industry (TAPPI) Standard:

T401m - 60 - Fiber Identification.
T403ts - 63 - Bursting Strength.
T405ts - - Paraffin in Paper.
T410os - 68 - Basis Weight.
T414ts - 65 - Tearing Resistance.

(Applications for copies should be addressed to the Technical Association of the Pulp and Paper Industry One Dunwoody Park, Atlanta, GA 30341.)

American Society for Testing and Materials (ASTM) Standards:

D87-66 - Melting Point of Paraffin Wax.

(Application for copies should be addressed to the American Society for Testing Materials, 1916 Race Street, Philadelphia, PA 19103.)

3. REQUIREMENTS

3.1 Material. The material used in the construction of the bags shall be of unbleached fibers, containing not less than 5 percent by weight of reclaimed fibers, having the weight and strength requirements specified herein. Reclaimed fibers are those reclaimed from solid waste or waste collected as a result of a manufacturing process but shall not include those materials generated from and reused within a plant as part of its own paper making process. The paper shall be sized to the extent that one drop of water (0.1 cubic centimeter) will not be completely absorbed into the paper when tested as specified in 4.4.2.

3.1.1 Type II. The nominal basis weight of paper shall be 35 pounds, tolerance plus or minus 5 percent (24 x 36-500).

3.1.2 Type II. The nominal basis weight of paper shall be 46 pounds, tolerance plus or minus 5 percent (24 x 36-500).

3.2 Wax. Bags, both type I and type II, shall be impregnated with not less than 4 pounds per 500 sheets (24 x 36) of parafin wax having a melting point of not less than 128°F. nor more than 134°F. The wax content and the melting point shall be determined as specified in 4.4.1.

3.3 Bursting strength.

3.3.1 Type I. The bursting strength of the waxed paper in the finished bag shall be not less than 20 points when tested as specified in 4.4.1.

3.3.2 Type II. The average bursting strength of the waxed paper in the finished bag shall be not less than 24 points when tested as specified in 4.4.1.

3.4 Tearing resistance.

3.4.1 Type I. The tearing resistance of the waxed paper in the finished bag shall be not less than 54 grams, machine direction, when tested as specified in 4.4.1.

3.4.2 Type II. The tearing resistance of the waxed paper in the finished bag shall be not less than 67 grams machine direction, when tested as specified in 4.4.1.

3.5 Construction.

3.5.1 Type I, square bags. The bags shall be so constructed that the face with lip, bottom and center seams or overlaps shall be to the rear. Seams and overlaps shall be sealed securely throughout their area, and shall be glued with a strong moistureproof adhesive (see 4.4.1.7).

3.5.1.1 Dimensions and capacities. The approximate dimensions of all type I bags are furnished in table I as a guide only. The dimensions of the bags as produced by different manufacturers may vary somewhat and be considered acceptable provided that such bags meet the applicable minimum cubic inch capacity as specified in table I.

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3.5.2 Type II, self-opening (automatic) bags. Type II bags shall be of the self-opening (automatic) type with a flat rectangular bottom, positioned so that the bottom is to the front or back. The front shall be the side opposite the seam. Seams and overlaps shall be sealed securely throughout their area, and shall be glued with a strong moistureproof adhesive (see 4.4.1.7).

TABLE I. Dimensions, capacity

Size Gallons	Face Width Inches	Tuck Width Inches	Bag Lengths Inches	Capacity Minimum Cu. inch
2	7-3/8	6-7/8	14	536
2	8	6	14-1/4	540
3	8	7-3/4	17-1/8	822
4	8	7-3/4	21-3/4	1108

Tolerance, plus or minus 1/4 inch.
One quart, dry measure, equals 67.20 cubic inches.

3.5.2.1 Dimensions and construction. The approximate dimensions and construction of the bag shall be as shown in table II and figure 3.

TABLE II. Dimensions, capacity

Face Width Inches	Tuck Width Inches	Bag Length Inches	Capacity Minimum Cu. Inch
8-1/2	5-1/2	18	842

Tolerance, plus or minus 1/4 inch.

3.6 Workmanship. Workmanship shall be first class throughout. Seams and overlaps on sides and bottoms of the bags shall be securely sealed throughout their area, and shall be glued with strong moistureproof adhesive. The bags shall be free from defects which detract from their appearance or which may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspection are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection. Sampling for inspection and acceptance shall be performed in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated.

4.2 Component and material inspection. In accordance with 4.1, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

4.3 Inspection of end item.

4.3.1 Examination of the end item. The end item shall be examined in accordance with the classification of defects, inspection levels, and acceptable quality levels (AQL's) set forth below. No more than five (5) bags, randomly selected, shall be drawn from any one shipping container from each lot of material for each type, size and capacity offered for acceptance for visual and dimensional characteristics. The lot size, for purposes of determining the sample size in accordance with MIL-STD-105, shall be expressed in units of unit packages of the specified material for examinations under 4.3.1.1 through 4.3.1.3 inclusive and in units of shipping containers for examination under 4.3.1.4.

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4.3.1.1 Examination of the end item for defects in appearance, construction, and workmanship. The sample unit for this examination shall be one bag.

Examine	Defect
Appearance	Surfaces not clean; presence of foreign or undigested matter, dirt spots, stains, discoloration, grease or oil spots.
	Tear, cut, puncture, split, holes (including pinholes), slits, breaks, wrinkles or sharp creases.
	Not smoothly and uniformly impregnated with wax on both sides; evidence of bare spots, lumps, blister, flaking, or streaked wax; presence of foreign matter imbedded in wax.
Construction	Not type specified; construction not in conformance with contractual requirements for the applicable type bag (see figs. 1, 2 and 3, as applicable).
	Seams poorly glued, do not adhere securely throughout their entire area; glue line not continuous throughout any joined area.
	Position of seams not as specified.
Workmanship	Folded edges not parallel or perpendicular to unfolded edges of bag.
	Bags stick together to the extent that opening causes tearing or injury to any surface.

4.3.1.2 Examination of the end item for dimensional defects. The sample unit for this examination shall be one bag. (NOTE: Method of measuring the bags and computing their respective capacities shall be performed in accordance with 4.4.3).

Examine	Defects
Dimensions:	
Type I and II	Length, face width or tuck width (as applicable) varies by more than plus or minus 1/4 inch from dimensions specified in table I or II for the applicable type and size bag.
Special sizes	Any dimension varying by more than plus or minus 1/4 inch in bottom width from size specified in contract or order.
Capacity	More than two percent below the contractual requirements for the specified type and size bag.

4.3.1.3 Examination of the end item for count. The sample unit for this examination shall be one unit package of bags. The average count of bags per package shall be not less than the specified quantity.

4.3.1.4 Examination of preparation for delivery. An examination shall be made to determine that packaging, packing and markings comply with the requirements of section 5 of this specification. The sample unit shall be one shipping container, fully packed, selected just prior to the closing operation. Shipping containers fully prepared for delivery shall be examined for closure defects.

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Examine	Defects
Packaging	Not in accordance with contract requirements; not unit wrapped and adequately secured as specified.
	Mixed type, size or capacity bags in unit package.
	Packaging material not as specified.
Packing (as applicable)	Not level specified; not in accordance with contract requirements.
	Arrangement or number of unit packages per shipping container not as specified.
	Liner material (as applicable) omitted, damaged or not type material specified.
	Container material not as specified; closures not accomplished by specified or required methods or materials.
Strapping (steel or tape as applicable)	Not properly placed lengthwise, girthwise or horizontally, or not applied in manner specified.
Markings	Interior or exterior markings (as applicable) illegible, incorrect, incomplete, omitted, or not in accordance with contract requirements.

4.3.1.5 Inspection levels and acceptable quality levels (AQL's) for examinations. The inspection levels for determining the sample size, and the acceptable quality levels (AQL's) expressed in defects per 100 units, shall be as follows:

Examination paragraph	Inspection level	AQL
4.3.1.1	S-4	4.0
4.3.1.2	S-2	2.5
4.3.1.3	S-2	---
4.3.1.4	S-2	4.0

4.3.2 Testing of the end item. The end item shall be tested for the applicable characteristics as indicated in table III for each lot presented for examination for each type, size and capacity bag. The sample unit for tests shall consist of five (5) bags. Five sample units, randomly selected throughout the lot, shall be tested with no evidence of failure to meet the specified requirements. NOTE: At the discretion of the contractor, with the exception of the seam strength characteristic, testing may be performed on samples of the treated paper prior to fabrication of the bags.

TABLE III

		1/			2/	3/	4/
Stock (dewaxed)							
Type I	3.1.1	4.4.1.1	-	-	2 (composite)	-	percent
Type II	3.1.2	4.4.1.1	-	-	2 (composite)	-	percent
Basic weight (dewaxed): (24 x 36 - 500)							
Type I	3.1.1	4.4.1.3	-	-	2 (composite)	-	0.1 pound
Type II	3.1.2	4.4.1.3	-	-	2 (composite)	-	0.1 pound
Wax content	3.2	4.4.1.2	-	-	Avg of 2 (composite)	-	0.1 pound
Melting point of wax	3.2	4.4.1.6	-	-	Avg of 3 (composite)	-	0.1°F.

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TABLE III (cont'd)

	1/	2/			3/		4/
Bursting strength:							
Type I	3.3.1	4.4.1.4	-	X	6 (3 on each side)	-	point
Type II	3.3.2	4.4.1.4	-	X	6 (3 on each side)	-	point
Tearing resistance: (Machine direction)							
Type I	3.4.1	4.4.1.5	-	X	Avg of 5	-	gram
Type II	3.4.2	4.4.1.5	-	X	Avg of 5	-	gram
Seam strength:							
Type I	3.5.1	4.4.1.7	X	-	2	X	-
Type II	3.5.2	4.4.1.7	X	-	2	X	-

NOTE: 1/ Tests as noted in 4.4.1, shall be performed in accordance with the referenced test methods of TAPPI or ASTM, as applicable, except that the stipulation as to the number of specimens to test, number of determinations to perform or the number of results to report shall be as indicated in this table.

2/ The composite sample shall consist of representative portions taken from each sample unit and combined into a single sample.

3/ If failure is indicated, report the description of failure.

4/ Test reports shall include all values on which results are based.

4.4 Tests.

4.4.1 Test procedures. Tests on materials used in the fabrication of the bags shall be conducted in accordance with the following methods as specified for the appropriate test.

4.4.1.1 Fiber identification. The fiber identification shall be determined in accordance with TAPPI T401m - 60. Prior to testing for fiber identification and quantitative determinations, extract the total wax from the specimen as specified in 4.4.1.2.

4.4.1.2 Total content of wax. A one square foot specimen of waxed paper cut from the sample bags selected shall be weighed and the weight noted. The wax content shall be determined in accordance with procedure specified in TAPPI Method T405ts. The content shall be reported as the weight of wax per ream.

4.4.1.3 Basis weight. The basis weight of the base stock paper shall be determined in accordance with Method TAPPI T410os-68 after removal of the wax in accordance with TAPPI Method T405ts.

4.4.1.4 Bursting strength. The bursting strength shall be determined in accordance with TAPPI T403ts-63. If specimen is tacky, it shall be dusted with talc on both sides, prior to conducting the bursting strength test, to prevent sample specimen from adhering to the instrument.

4.4.1.5 Tearing resistance. The tearing resistance, machine direction, shall be determined in accordance with TAPPI T414ts-65.

4.4.1.6 Melting point of wax. The melting point of the wax, as specified in 3.2 shall be determined as follows: Scrape a small quantity of surface wax from the paper using a sharp edge such as a microscope slide or a razor blade, using care not to contaminate the wax with paper fibers. Place approximately one tenth (0.1) gram of the wax between micro cover glasses. Place this assembly upon the platform of a hot stage melting point apparatus (Fisher-Johns Melting Point Apparatus or equal). Adjust the heating rate of the apparatus to 3° ± 1°F. per minute and observe the temperature at which the sample completely liquefies. This temperature is recorded as a melting point determination of the material. Perform melting point determinations upon the same 0.1 gram sample until not less than three (3) determinations differing by not more than 0.5°F. are obtained. Record the average of these determinations as one melting point test result. In case of dispute, a referee retest shall be performed upon a representative sample of material from the first using ASTM Method D87-66.

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4.4.1.7 Moistureproofness of adhesive. Two sections of the glued seam of the bag, at least 3 inches long and of such width that there shall be at least 1 inch of material on each side of the finished seam, shall be taken from each bag. Each section shall be completely immersed in water at room temperature for a period of not less than 6 hours. At the end of that time, the bag seams shall not have separated, nor shall they be capable of being separated without pulling the fibers of the paper.

4.4.1.8 Sizing test. After conditioning to equilibrium, the specimen of paper shall be placed on a flat, dry and clean surface. A drop of water (approximately 0.1 cubic centimeter) shall be dropped on the specimen from a height of not more than 3 inches above the specimen. A watch glass shall be placed over the drop of water immediately after the drop has contacted the paper. After 20 minutes the area of the paper contacted by the drop shall show partial retention of the drop as evidenced by reflectivity.

4.4.2 Methods of measuring the bags and computing their capacities.

4.4.2.1 Type I, square bags. The position of the bag shall be such that the face with the lip is to the rear. Measurements to be made on the finished bag are indicated by the dimensional lines shown in figure 1, which do not include the lip length as part of the length measurement.

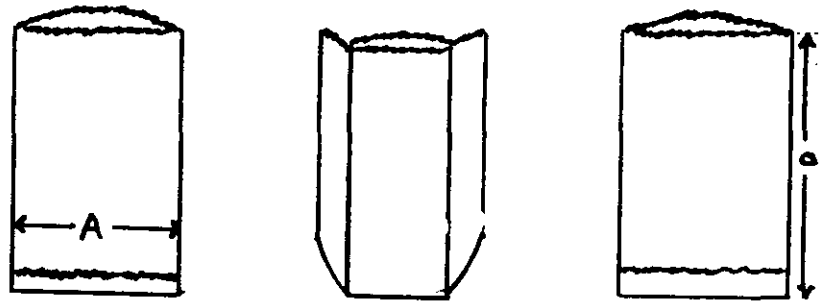
4.4.2.1.1 Face width, A. Measure across the front of the folded bag just above the top of the bottom fold.

4.4.2.1.2 Tuck width, C. Spread the bag so that the tucks normally behind the face are fully extended outward. Press the bag flat and measure full width as close to the bottom fold as possible. Subtracting the face width from this measurement gives dimension C.

4.4.2.1.3 Bag length, D. Measure along center of the front face of the bag, from the lower bottom fold to the top of the serrated edge of the front face.

4.4.2.1.4 Cubic capacity. Cubic capacity $\sqrt{A \times C \times (D - 1/2C)}$ is the product of the face width, tuck width, and bag length, minus one-half the tuck width.

4.4.2.2 Type II, self-opening (automatic) bags. The position of the bag shall be such that the bottom is to the front. Measurements to be made on the finished bags are indicated by the dimensional lines shown in figure 2.



1/2 Perimeter

FIGURE 1

4.4.2.2.1 Face width, A. Measure across the front of the folded bag just above the top of the bottom fold.

4.4.2.2.2 Tuck width, C. Spread the bag so that the tucks normally behind the face are fully extended outward. Press the bag flat and measure full width as close to the bottom fold as possible. Subtracting the face width from this measurement gives dimension C.

4.4.2.2.3 Bag length, D. Measure along center of the front face of the bag, from the lower bottom fold to the top of the serrated edge.

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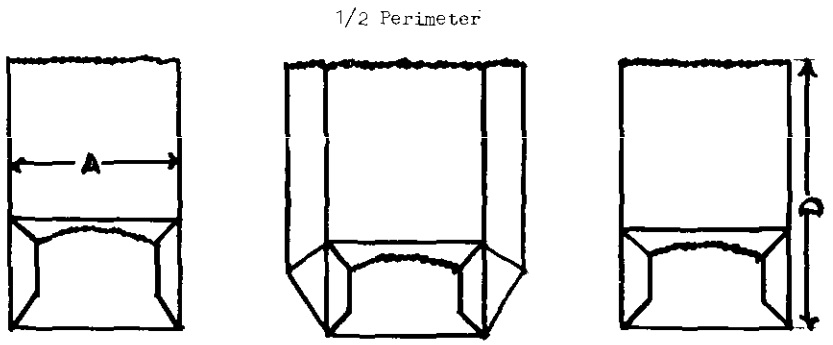


FIGURE 2

4.4.2.2.4 Cubic capacity. Cubic capacity (A x C x D) is the product of the face width, bottom width and bag length.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B or C as specified (see 6.1).

5.1.1 Level A. One hundred (100) bags shall be cross tied with twine into bundles.

5.1.2 Level B. Fifty bags (50) shall be cross tied with twine into bundles.

5.1.3 Level C. Bags shall be packaged to afford adequate protection against deterioration and physical damage during shipment from the supply source to the first receiving activity. The supplier may use his standard practice when it meets this requirement.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.1).

5.2.1 Level A. The bags shall be packed in close-fitting boxes conforming to PPP-B-585, class 3, PPP-B-591, class II, PPP-B-601, type III, PPP-B-621, class 2, PPP-B-636, V3c, or PPP-B-640, class 2, grade A. The boxes shall be closed and strapped in accordance with the applicable specification or appendix thereto. The gross weight of the triple wall fiberboard box and the wood boxes shall not exceed approximately 200 pounds. The gross weight of the PPP-B-636 fiberboard box shall not exceed the weight limitations of the box specification.

5.2.2 Level B. Twelve (12) unit packages of bags, in a reversed end order, shall be packed in a close-fitting box conforming to PPP-B-636, class domestic. Closure shall be method I of the appendix to the box specification.

5.2.2.1 Military agencies. Closures of fiberboard boxes shall be in accordance with method II of the appendix to PPP-B-636.

5.2.3 Level C. Bags packaged as specified in 5.1 shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with rules and regulations of carriers applicable to the mode of transportation.

5.3 Standard pack (civil agency procurements). The standard pack for civil agency procurements will be as specified in paragraphs 5.1.1 and 5.2.2 (see 6.2).

5.4 Marking.

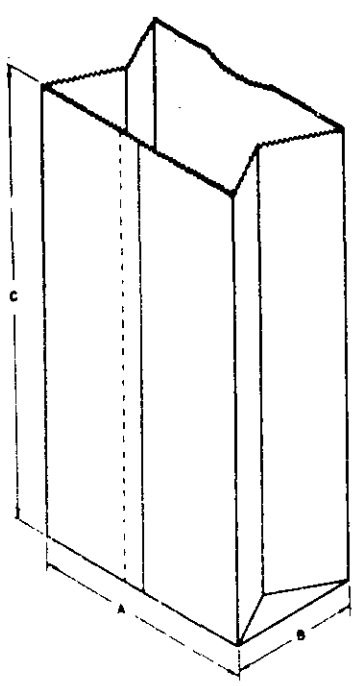
5.4.1 Civil agencies. In addition to markings required by the contract or order, the package and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.4.2 Military agencies. In addition to markings required by the contract or order, the packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

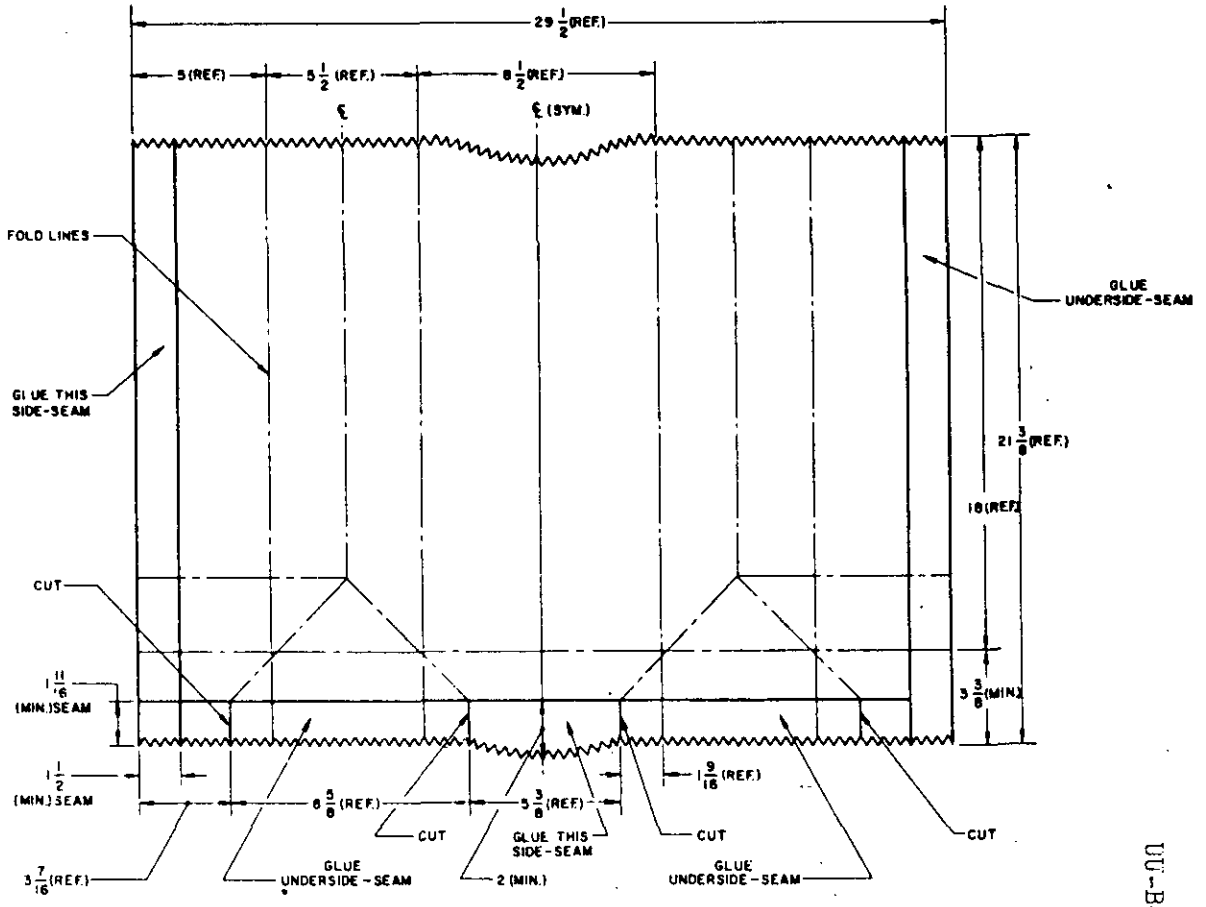
6.1 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents.

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ASSEMBLY

FACE WIDTH	TUCK WIDTH	BAG LENGTH
A	B	C
$8\frac{1}{2} \pm \frac{1}{4}$	$5\frac{1}{2} \pm \frac{1}{4}$	$18 \pm \frac{1}{4}$



UNFOLDED VIEW

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FIG. 3 – TYPE II

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(a) Title, number and date of this specification.
(b) Type and size required (see 1.2.1 and 1.2.2).
(c) Level of packaging and packing required (see 5.1 and 5.2).
(d) Inspection responsibility, methods and tests (see section 4).

6.2 Standard pack for civil agencies. The standard pack requirements in 5.3 are intended for use in procurements of stores stock replenishments. Procuring officers should use the standard pack requirements when it is known that the material will be shipped from a supplier to a domestic warehouse, supply depot or intermediate storage point for temporary storage, subsequent issue or shipment to eventual user.

MILITARY INTERESTS:

Army - M
Navy - MD
Air Force

Preparing activity:

GSA-FSS

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